

1

## Build the graph

rollout  $\pi_t$ 

k-means · N nodes

observed graph  $\hat{F}$  $\hat{F}$  $A$  $C_k$ 

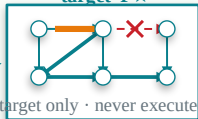
$$\langle C_k, F \rangle \leq b_k = B_k / L$$

$\bar{J}_k$

2

## Project + price

same transition graph, constrained once

observed  $\hat{F}$ target  $F^\star$ 

target only · never executed

$$\max_F \langle A, F \rangle - \varepsilon \text{KL}(F \| \hat{F})$$

budget occupancy

k=1



$$\lambda^\star_1 > 0$$

k=2



$$\lambda^\star_2 = 0$$

$$\lambda_k = \bar{\lambda}_k^\star + \beta_k$$

+

clip + smooth

$$\bar{J}_k - B_k \rightarrow \beta_k$$

next rollout

3

## Priced actor

GAE from rollout  
reward  $A_r$  · cost  $A_c$ 

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO  $\rightarrow \pi_{t+1}$ 